CS4085 DEEP LEARNING · FINAL RESEARCH PROJECT

HireMind: Efficient CV–Job Fit Classification through Distillation and Quantization

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Code, data splits and results on GitHub

Problem and data

Recruiters compare every CV with the job by hand: slow.

Large AI models need a GPU or send CVs to an outside service.

Goal: a small model on a normal CPU that keeps CVs private.

Labels: No Fit · Potential Fit · Good Fit. Jobs never repeat across splits (474 CVs do). Inputs cut to 256 tokens.

4,993	1,162	1,759
training	validation	test pairs

Research question and contributions

Can teacher-guided training improve a compact CV–job classifier, and can quantization cut its cost while keeping useful performance?

1. **Deep learning: distillation.** A BERT-base teacher guides the small student.
   H1: M1 gains ≥ +1 macro-F1 point over M0

2. **Efficiency: quantization.** M1's weights are stored in 8 bits.
   H2: ≤ 40% of M1's storage, ≤ 1 point loss

Model architecture: M0, M1 and M2

Teacher: BERT-base
fine-tuned on training data

soft labels (M1 only), T = 2

CV + job
one pair, 256 tokens → **BERT-Tiny student**
2 layers · width 128 · 4.39 M params → **3-class head**
No · Potential · Good

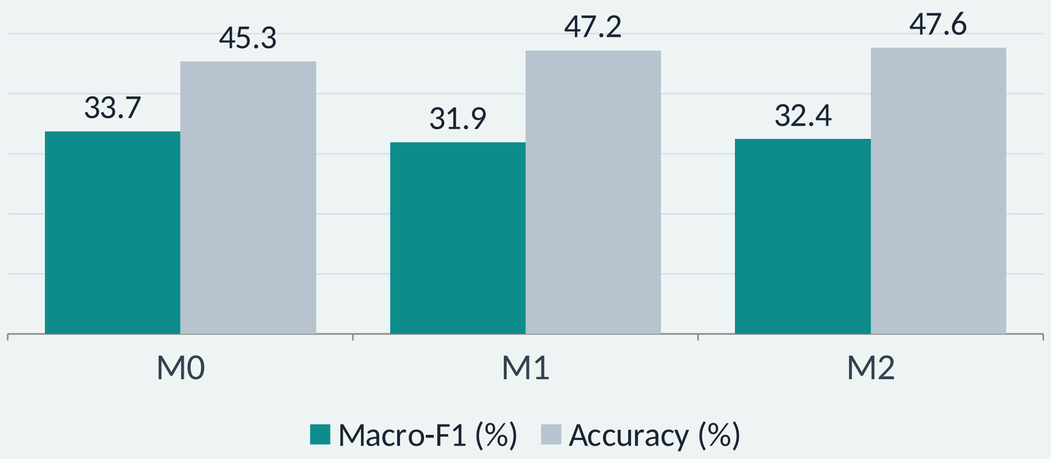
Loss for M1: $L = 0.5 \cdot CE + 0.5 \cdot T^2 \cdot KL(\text{teacher} \parallel \text{student})$

M2 = M1 with INT8 Linear layers and 8-bit embeddings, no retraining. Same student, data and budget for M0 and M1.

Three trained versions

M0 CE only → **M1** CE + KL → **M2** M1 in 8-bit

Held-out test results (3 seeds)



M0 is best on macro-F1 (33.70%). Accuracy rises from M0 to M2, but class-balanced macro-F1 falls. Seed SD is 3.9–5.2 points.

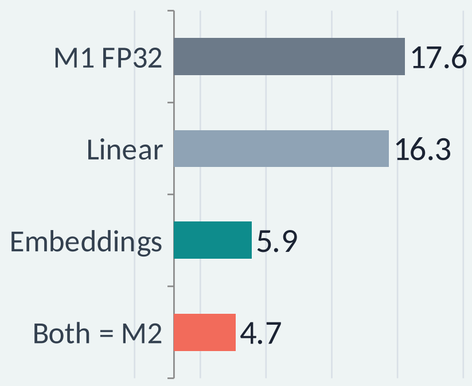
Efficiency and ablation

−73.4% **saved size**
17.56 → 4.67 MB

+13.5% **CPU latency**
5.56 → 6.31 ms

Smaller is not faster: parameters (4.39 M) and MACs (0.134 G) do not change.

Quantization ablation: size (MB)



Embeddings give most of the saving; validation macro-F1 barely moves (37.28 → 36.96). Temperature 1 vs 2: no consistent effect.

Error analysis

	No	Pot.	Good
No	798	13	46
Pot.	356	39	49
Good	419	32	**7**

M2, seed 44: only 7 of 458 Good Fit pairs found; 419 called No Fit. M1 already failed at this seed.

Hypotheses

H1: distillation +1 point	**Not met (−1.82)**
H2: storage ≤ 40%	**Met (26.6%)**
H2: loss ≤ 1 point	**Met (+0.51)**
H2: keeps M1's gain	**Not testable**

Findings and limits

- Baseline wins; distillation did not help.
- Quantization: smaller, not faster.
- Limits: 3 seeds, 1 teacher, 256 tokens, familiar CVs, one CPU.
- Senior 2: advisory signal only; HR decides.

AI tools (ChatGPT, Claude) supported code debugging and editing; all results come from the team's own runs. · github.com/RAWANKHOJAH/HireMind-DL